P425/2

PRINCIPAL

MATHEMATICS

Paper 2

AUGUST 2026

3 hours

UNNASE MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

Paper 2

3 HOURS

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of **two** Sections; **A** and **B**. It has **five** examination Items.*

*Section **A** consists of **one compulsory** item.*

*Section **B** has **two** parts; **I** and **II** each with **two** optional items. Respond to only **two** items from Section **B**, choosing **one** item from **each** of the parts, **I** and **II**..*

*Respond to **three** items in all.*

*Any additional item(s) responded will **not** be scored.*

Begin each item on a fresh page.

All *necessary working* **must** *be shown clearly*

Graph paper is provided.

Where necessary, use g = 9.8 ms^{-2}.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

CamScanner

SECTION A:

*The item in this section is **compulsory***

ITEM 1

A construction company has been awarded a contract to install a drainage channel alongside a newly constructed straight road. For planning purposes, the engineers represent the drainage channel by the curve $y = x^3 - 7x + 6$, while the road is represented by the x-axis, $y = 0$.

To determine the points where the drainage channel intersects the road, the engineers intend to sketch the graph of the curve and estimate the three points of intersection within the interval $-4 \leq x \leq 3$.

In addition, the company plans to construct public washrooms in the region enclosed by the drainage channel and the road between $x = 2$ and $x = 3$. To assess whether the available land is sufficient, the engineers will estimate the enclosed area with six ordinates. If the estimated area is less than 4 square units, the site will be considered too small for the proposed washrooms.

The company also intends to purchase cylindrical concrete culverts for the drainage system. The quoted dimensions of each culvert are; radius of 80.45 cm and a height of 120.2 cm correct to the nearest decimal place. Before placing the order, the engineers wish to verify the reliability of these measurements. They will determine the percentage error in the volume of the culvert and proceed with the purchase only if the percentage error is less than 1.0%.

Task.

Help the construction company make informed decisions using the information above.

SECTION B:

*Answer only **two** items from this section, choosing **one** item from each of the parts **I** and **II***

Part I

Item 2

A roadside vendor operating along the Kampala–Jinja Highway near Mabira Forest sells roasted chicken to travelers and commuters every day. To meet the daily demand, she stocks 100 pieces of chicken each morning. After keeping sales records for several weeks, she established the following probability distribution for the number of pieces sold each day.

Number of pieces of chicken sold	0	20	40	60	80	100
Probability of selling	0.02	0.16	0.22	0.26	0.20	0.14

CamScanner

The vendor intends to use this information to estimate the expected number of pieces of chicken sold per day and the standard deviation of the daily sales. These statistics will help her plan future stock levels and minimize unnecessary losses due to unsold products.

Each piece of chicken is purchased from suppliers at UGX 3,000 and sold to customers at UGX 6,000. Since any unsold pieces are consumed by her family and therefore have no resale value, she wishes to determine her expected daily profit. If the expected daily profit is less than UGX 50,000, she intends to apply for a loan from a local money lender to increase her business capital and expand her daily stock.

To ensure that she sells high-quality products, the vendor purchases chicken from two wholesalers, A and B. The probability that she purchases her stock from wholesaler A is ¼, while the probability of obtaining good-quality chicken is $^2/_3$ when buying from wholesaler A and $^1/_3$ when buying from wholesaler B. She wishes to determine the overall probability that a randomly selected piece of chicken from her stock is of good quality. If this probability is less than $^2/_5$, she plans to look for alternative suppliers with more reliable quality.

On one particular day, the vendor randomly selects five pieces of chicken from her stock for inspection. Assuming that each selected piece is equally likely to have been supplied by either wholesaler according to the stated probabilities, she wishes to determine the probability that exactly three of the five pieces were supplied by wholesaler A.

Task.

Help the vendor obtain the necessary information to guide informed decision making.

ITEM 3.

A secondary school administration carried out an evaluation of students' performance in Economics to determine whether there had been any improvement in academic achievement over the course of the term. To do this, the administration compared the Beginning of Term (BOT) examination scores of ten students with their corresponding End of Term (EOT) examination scores. The results are shown in the table below.

BOT(x)	70	60	58	45	80	89	45	72	58	51
EOT(y)	65	53	55	47	75	86	58	70	65	44

The administration wishes to illustrate the relationship between the BOT and EOT scores using an appropriate graph and determine whether a statistically significant relationship exists between the two sets of scores.

CamScanner

If the relationship is found to be significant at the 5% level of significance, the administration intends to organize an academic review meeting with the Economics teachers to discuss students' performance.
In addition, the Head of the Economics Department has been asked to prepare a statistical summary of the End of Term examination results for the entire class. The grouped frequency distribution of the EOT marks is presented below.

Marks	35-<45	-<50	-<60	-<65	-<75	-<80	-<95
Number of students	4	13	25	30	20	15	3

The administration would like to determine the modal mark for the End of Term examination. If the modal mark is less than 55, a meeting will be arranged with the students to discuss strategies for improving their academic performance.

Furthermore, the Head of the Economics Department has advised that the End of Term examination scores are approximately normally distributed. Based on this assumption, the administration wishes to identify the number of students who scored more than 85 marks, as these students will be appointed as leaders of the various Economics discussion groups.

Task.
Help the administration obtain the information needed to make informed decisions.

Part II

ITEM 4

A football club is conducting a pre-season fitness assessment to prepare its players for an upcoming tournament. As part of the strength-training programme, one of the players, who has a mass of 80 kg, performs a suspension exercise by hanging motionless from two training ropes attached to an overhead support. The ropes are inclined at 20° and 40° to the vertical.

According to the manufacturer's safety specifications, each rope can safely withstand a maximum tension of 300 N. Before continuing with the exercise, the coaching staff wishes to determine the tension in each rope and decide whether the ropes are safe for use or should be replaced. During the same training session, the player practices shooting accuracy by kicking a football from level ground with an initial speed of 30 m s^{-1}. He aims to land the ball at a teammate positioned 45 m away on the same

CamScanner

horizontal pitch. To improve his shooting technique, the coach asks him to determine the possible angles of projection that will enable the ball to reach the target.

To conclude the training session, the player completes an endurance drill by running two consecutive distances of 500 m. He covers the first 500 m in 1 minute and the second 500 m in 3 minutes. Assuming that his speed decreases with uniform retardation throughout the drill, the coach wishes to estimate the additional distance the player would run before eventually coming to rest.

Task:
Help the coaching staff analyse the player's training performance and make informed decisions.

ITEM 5.

The Police Traffic Control Unit is monitoring vehicle movement along the busy Kampala–Jinja Highway due to a recent increase in road accidents. To enhance safety, a real-time vector-based tracking system has been installed to monitor positions and velocities of moving vehicles in three-dimensional space.

On a particular day;

- A trailer departs from Kampala at 11:00 a.m, position $(2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})$km with constant velocity $(2\mathbf{i}+2\mathbf{j}+3\mathbf{k})$km/h
- A bus departs from Jinja at 12:00 noon from position, $(7\mathbf{i} + 5\mathbf{j} - 2\mathbf{k})$km with constant velocity $(-3\mathbf{i} - 8\mathbf{j} + 18\mathbf{k})$km/h

Due to the high traffic density, the control center must determine whether the two vehicles are on a collision course and, if so, identify the exact point of collision in order to dispatch emergency services.

According to new traffic regulations, all accident vehicles must be towed separately to a nearby police station for forensic investigation.

In one such incident, a trailer of mass 5 tonnes is towed using a recovery vehicle of mass 1.5 tonnes with a driving force of 60 kN up a rough road inclined at 3 in 5, coefficient of friction between the vehicles and the road is 0.25

Tasks

(a) Use the information above to determine whether a collision occurs and find the time and position of the collision

(b) Calculate the tension in the tow bar between the trailer and the recovery vehicle

*****End*****

CamScanner